Assignment 2 Report

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Overview

This report covers all following tasks: 1. Training word embeddings with SVD and Word2Vec (Skip-Gram + negative sampling) 2. Geometric/ethical analysis of SVD, Word2Vec, and pre-trained GloVe embeddings 3. POS tagging with MLP using the three embedding variants

Task 1: Training Word Vectors

1.1 SVD Embeddings (`svd_embeddings.py`)

Method

- Corpus: Brown (`nltk.corpus.brown`)
- Preprocessing:
  - lowercase
  - keep only alphabetic tokens (`token.isalpha()`)
- Build co-occurrence matrix with symmetric context window
- Convert co-occurrence matrix to PPMI
- Apply truncated SVD (`svds`) and use `U @ diag(s)` as embeddings
- Save artifacts to `embeddings/svd.pt`

Hyperparameters searched in code

- `min_count`: `[4]`
- `context_window`: `[2, 4]`
- `k_components`: `[100, 200, 300]`

Best saved hyperparameters (from `embeddings/svd.pt`)

- `min_count=4`
- `context_window=4`
- `k_components=300`
- Relative reconstruction error: `0.9153`

Justification

- `min_count=4` removes very rare words, reducing noise and making co-occurrence estimates more reliable.
- `context_window=4` gives broader semantic context than window 2, which usually helps global/topic-level relations in count-based models.
- `k=300` is a common embedding size and preserves more information than 100/200 while still compressing the matrix.
- Selection criterion is reconstruction error on the PPMI matrix.

1.2 Neural Embeddings: Word2Vec Skip-Gram (`word2vec.py`)

Method

- Model: Skip-Gram with negative sampling
- Negative sampling distribution: unigram^0.75
- Loss: `BCEWithLogitsLoss` over positive + negative pairs
- Optimizer: Adam
- Early stopping with patience 3

Hyperparameter search space

- `embedding_dim`: `[100, 200, 300]`
- `window_size`: `[2, 4]`
- `negative_k`: `[5, 10]`
- `learning_rate`: `[0.001, 0.003]`
- `batch_size`: `[256, 512]`
- Total configs: `48`

Best config

- `embedding_dim=100`
- `window_size=4`
- `negative_k=10`
- `learning_rate=0.003`
- `batch_size=512`
- `epochs_run=6` (early stopped)
- `best_val_loss=1.4230`

Justification

- `window_size=4` consistently appears in top configurations, indicating larger context helps this corpus.
- `negative_k=10` slightly improves discrimination by providing more contrastive negatives.
- `lr=0.003` converges faster and reached lower validation loss before early stopping.
- `batch_size=512` stabilizes updates and was present in top-performing settings.
- `embedding_dim=100` winning over larger dims suggests better bias-variance tradeoff for this data/training budget.

Task 2: Are the Embeddings Fishy?

2.1 Analogy test (Top-5) for required three analogies

Paris : France :: Delhi : ?

- SVD: `treaty, britain, germany, holland, europe`

- Word2Vec: `pratt, airports, woonsocket, prevot, maximization`
- GloVe: `india, pakistan, indian, punjab, kashmir`

King : Man :: Queen : ?

- SVD: `a, he, young, him, woman`
- Word2Vec: `cement, rich, ellen, mary, club`
- GloVe: `woman, girl, person, she, lady`

Swim : Swimming :: Run : ?

- SVD: `hit, gamut, ritchie, poked, triple`
- Word2Vec: `supports, liberalism, spirits, offering, little`
- GloVe: `running, runs, three, two, four`

Interpretation

- GloVe shows strongest semantic/syntactic consistency on all three required analogies.
- SVD and Word2Vec capture some local associations but often miss the intended relation.
- The likely reason is corpus size/domain (Brown) and limited training setup compared with large pre-trained corpora.

2.2 Bias check (GloVe only)

Profession	cos(profession, man)	cos(profession, woman)
doctor	0.4012	0.4691
nurse	0.2373	0.4496
homemaker	0.0529	0.2857

Interpretation

- In this snapshot, all three professions are closer to `woman` than `man`.
- This indicates the embedding reflects gendered association patterns from training data, matching known embedding bias behavior.

Task 3: POS Tagging with MLP

3.1 Data and setup

- Brown corpus with universal tagset (`tagset='universal'`)
- Preprocessing: lowercase + alphabetic tokens only
- Split: 80% train / 10% val / 10% test (seed 42)
- Input representation: sliding window + concatenation
- Boundary handling: PAD index with zero vector

- Embeddings kept frozen during MLP training

Universal tags are a simplified POS set used for consistent evaluation across corpora. In this report, tags like `NOUN`, `VERB`, `ADJ`, `ADV`, `PRON`, `DET`, and `ADP` capture core grammatical roles, while `PRT`, `CONJ`, `NUM`, and `X` handle particles, conjunctions, numbers, and miscellaneous tokens.

3.2 MLP architecture

- Input: `(2C+1) * emb_dim`
- Hidden layer 1: `hidden_size`
- Hidden layer 2: `hidden_size // 2`
- Activations: ReLU
- Regularization: Dropout `0.3`
- Output: logits over POS tags

3.3 POS hyperparameter tuning

Search space (per embedding type): - `window_size`: `[1, 2]` - `hidden_size`: `[256, 512]` - `lr`: `[0.001, 0.003]` - `batch_size`: `[256, 512]` - Total: `16` per embedding, `48` overall

Best per embedding: - GloVe: `window=1, hidden=512, lr=0.001, batch=256, dropout=0.3, best_val_loss=0.0860` - SkipGram: `window=1, hidden=512, lr=0.001, batch=512, dropout=0.3, best_val_loss=0.1187` - SVD: `window=1, hidden=512, lr=0.001, batch=512, dropout=0.3, best_val_loss=0.1461`

3.4 Test performance

Embedding	Accuracy	Macro-F1
GloVe	0.9743	0.9409
SkipGram	0.9629	0.9106
SVD	0.9527	0.8941

3.5 Confusion matrix (best model: GloVe)

The confusion matrix is strongly diagonal, which indicates the model predicts most tags correctly. The largest off-diagonal cells appear in linguistically close categories, especially `ADJ <-> NOUN`, `PRT -> ADP`, and `VERB -> NOUN`, showing the model mostly fails on boundary/ambiguity cases rather than random errors.

Main confusion trends: - `ADJ <-> NOUN` - `PRT -> ADP` - `VERB -> NOUN`

These are linguistically plausible confusions under short context windows.

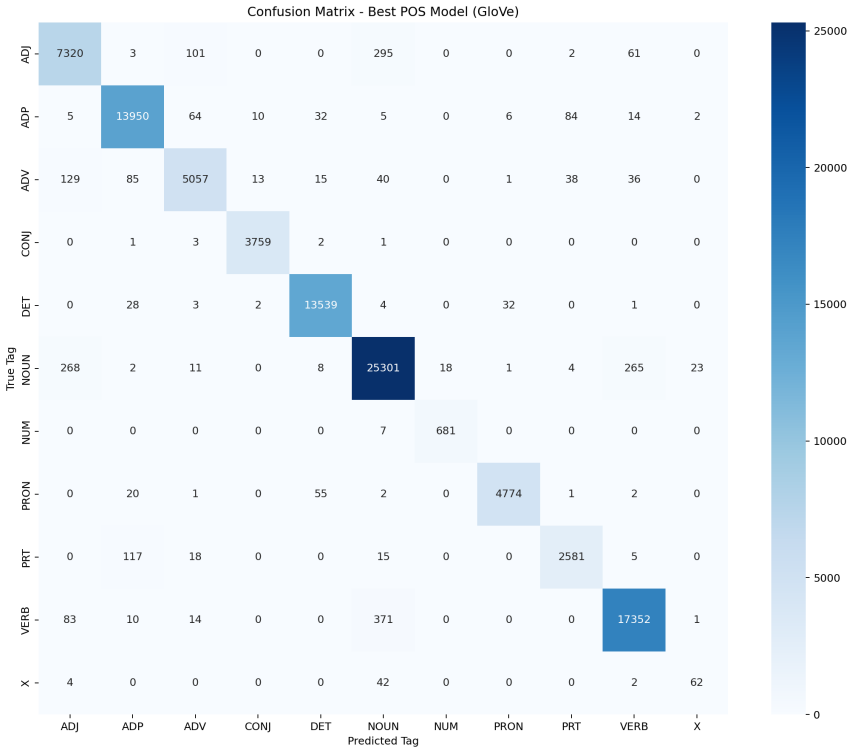

Figure 1: Confusion Matrix - Best POS Model (GloVe)

Task 4: Analysis and Report

4.1 Error Analysis (All POS Models)

4.1.1 GloVe model examples

1. `and dauntless by greentree adios in`

- `dauntless`: ADJ -> NOUN
- `adios`: NOUN -> ADV
- Reason: adjective/noun ambiguity and sparse context

2. `these responses are explicable in terms of characteristics inherent in the crisis`

- `characteristics`: NOUN -> ADJ
- `inherent`: ADJ -> NOUN
- Reason: nominal/adjectival ambiguity

3. `... called him in as a private lawyer ...`

- `in`: PRT -> ADP
- Reason: particle/preposition ambiguity

4.1.2 SkipGram model examples

1. `merciful god julia`

- `merciful`: ADJ -> NOUN (OOV)
- Reason: unseen token for this vocabulary

2. `... a chartered electrical engineer ...`

- `chartered`: VERB -> NOUN
- Reason: verb/noun ambiguity

3. `... set about maximizing ...`

- `about`: ADV -> ADP
- `maximizing`: VERB -> NOUN
- Reason: function-word ambiguity + noun bias

4.1.3 SVD model examples

1. `merciful god julia`

- `merciful`: ADJ -> NOUN (OOV)
- Reason: unseen/rare lexical item

2. `... an unwillingness to generalize and to search ...`

- `to`: PRT -> ADP
- `generalize`: VERB -> NOUN
- `search`: VERB -> NOUN

- Reason: infinitival marker ambiguity and verbal noun-like contexts

3. `these responses are explicable in terms of characteristics inherent in the crisis`

- `explicable`: ADJ -> NOUN
- Reason: adjective/noun confusion

4.2 Embedding Comparison Summary

- Pre-trained GloVe significantly outperformed SVD and SkipGram in both analogy quality and POS tagging.
- SkipGram is clearly better than SVD on downstream POS metrics.
- SVD still captures useful structure, but it is weaker for fine-grained semantic and syntactic regularities in this setup.

Conclusion

This assignment shows a consistent ranking across both intrinsic and extrinsic evaluations: `GloVe > SkipGram > SVD`. Pre-trained embeddings provided better semantic geometry (analogy quality), transferred better to POS tagging, and produced the highest test accuracy and macro-F1. Error analysis indicates most failures come from natural linguistic ambiguity (e.g., `ADJ/NOUN`, `PRT/ADP`, `VERB/NOUN`) and OOV effects, suggesting future gains would likely come from richer context models and subword-aware embeddings.